

SOC Game

The expressiveness test, and the operations vocabulary the graph would have to hold

The page is a title with nothing under it. What belongs under it is a question I have already put to myself twice, once on the cluster overview and once on the CHART software page, in almost the same words both times.

“CHART is only a first step to capture the syntax of human-agent teamwork. Its completeness and succinctness remain to be tested. Can it express the teamwork in a SOC center or a incident response team? Can we prescribe, monitor, and diagnose human-agent teamwork represented with CHART?”

website, Human-AI Complementarity, on what remains

“Can it express how a security operations centre or an incident-response team actually works?”

website, CHART artifact, What the next version has to answer

The assistantship posting says which direction the work runs in, and it runs in both.

“encoding real team structures to find what the graph cannot say, and extending the platform so that the structures it can express actually run”

website, Work With Me, Testing what CHART can express

So the SOC Game is not a fifth platform. It is the test that decides whether the fourth one is a formalism or a description, and the honest statement of it today is that the graph has been shown to run and has not been shown to hold anything real. This file is short because of that. It carries the centre’s own vocabulary, which is what the test would have to encode, and it names the three places where I already expect the encoding to fail.

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1 Active

1.1 The test

SOC Game

Definition: An expressiveness test for CHART, in which the division of authority and information in a real security operations centre is written down as a dependency graph, run as a session, and then checked against how the centre actually works.

Distinguishes it from: The three studies that already exist in this cluster. Team Defense Game, FriendOrFoe and CHART all fix a small team and vary its structure to see what the variation does. Here the structure is given by the world and the question is whether the notation can hold it. The dependent variable is the notation, not the team.

Operational test: Take a real shift's coordination pattern and try to write it as nodes and typed edges. Every part that needs a footnote in prose instead of an edge is a finding, and the count of those footnotes is the result.

In my project: It sits on the site as a heading with an empty page under it, and it is honest that it does. Nothing about the design is settled. What is settled is the question, which is stated twice in my own words, and the direction of the work, which is stated once in the posting.

expressiveness

Definition: The property of a notation that decides which real arrangements can be written in it at all, before any question of whether writing them is convenient.

Distinguishes it from: Coverage. Coverage is how much of a corpus a notation happens to handle. Expressiveness is what it can handle in principle, and the way to find its limit is to reach for the case it was not designed against.

Operational test: Find one arrangement the notation cannot write. That is worth more than a hundred it can, because it names the missing construct rather than confirming the present ones.

In my project: The five dependency types were each drawn from a coordination pattern in security operations, so the graph is expected to fit. That expectation is the reason the test has to be run against a centre nobody designed the types from.

completeness and succinctness

Definition: The two ways a notation can fail. Incomplete, when a real arrangement has no expression in it. Not succinct, when the arrangement has an expression but only as a large and unreadable one.

Distinguishes it from: Each other, which matters because the repairs differ. Incompleteness is repaired by a new construct. Loss of succinctness is repaired by an abbreviation over constructs that already exist, and adding a construct for it makes the notation worse.

Operational test: Write the arrangement. If you cannot, the notation is incomplete. If you can and the graph has forty edges where the centre has one rule, it is not succinct.

In my project: Both words are mine and both are on the overview page in the same sentence. Keeping them apart is what stops the test from ending in a request for a sixth dependency type.

prescribe, monitor, diagnose

“Can teamwork written in CHART be prescribed and enforced as well as recorded?”

website, CHART artifact, What the next version has to answer

“Can it be monitored during a session and diagnosed afterwards, so a structural fault surfaces while it is still correctable?”

website, CHART artifact, What the next version has to answer

Definition: The three things a written structure would have to support beyond being written: enforcing it on what may happen, watching conformance to it while a session runs, and attributing a failure to it afterwards.

Distinguishes it from: Recording, which CHART already does. A recorded structure tells you what the team was configured to do. A prescribed one refuses what falls outside it, a monitored one says while the shift is running that the team has drifted out of it, and a diagnosed one says afterwards which edge was the fault.

Operational test: Ask when the fault becomes visible. Afterwards is diagnosis. During, in time to change something, is monitoring, and only the second is any use to the people on shift.

In my project: The three verbs come from the overview page and the three questions from the software page, and they are the same list. Monitoring is the one nothing in the platform currently does, because everything in it is built to be read after the session.

1.2 The centre’s own vocabulary

security operations centre

Definition: The standing team that monitors an organization’s systems for security events, decides which of them matter, and drives the response, staffed in shifts and running continuously.

Distinguishes it from: An incident response team, which assembles around a confirmed incident and dissolves when it closes. The centre is permanent and its work is mostly the events that turn out to be nothing. My own sentence names both because they are two different structures, and a graph that fits one need not fit the other.

Operational test: Ask what the team does on a day with no incident. A centre still has a full shift of work. A response team has none.

In my project: This is the setting the whole cluster claims to be about, and it is the one setting none of my four studies has actually been run in. The five dependency types were drawn from it at second hand.

alert

Definition: A single machine-generated claim that something observed may be a security event, emitted by a detector into a queue for a person to judge.

Distinguishes it from: An incident, which is what an alert becomes once a person has confirmed it and scoped it. The gap between the two counts is the whole workload problem: practitioner surveys put a median team near a thousand alerts a day, and a substantial share are never looked

at by anyone.

Operational test: Ask who or what asserted it. A detector asserted an alert. A person asserted an incident.

In my project: The FriendOrFoe task compresses alert judgement into a binary malicious or benign call on twenty events. That is the laboratory form. The centre's form has a queue behind it that never empties, and the queue is the part the laboratory form drops.

triage

Definition: Deciding, per alert and under time pressure, whether it is a genuine threat, a false positive, or something that has to go to somebody with more authority.

Distinguishes it from: Investigation, which is what happens after triage sends something onward. Triage is bounded by the queue and optimizes throughput. Investigation is bounded by the incident and optimizes certainty.

Operational test: Ask what the analyst is allowed to spend. In triage the budget is minutes and the queue keeps growing while it is spent.

In my project: This is the task an AI teammate is most often proposed for, and it is the task where over-reliance is cheapest, because agreeing is fast and checking is not.

tier

Definition: The centre's division of its own labour by decision authority: the first tier triages against written procedure, the second investigates what the first sends up and may take containment actions the first may not, the third hunts for what no detector raised.

Distinguishes it from: Seniority. Tiers are defined by which actions a person may take without asking, and that is a graph property rather than a personnel one, which is what makes it the part of a centre most likely to be expressible.

Operational test: Name an action and ask who can take it without asking. The answer partitions the staff, and the partition is the tiering.

In my project: This is the most direct fit between the centre and the notation. Tiering is a chain of control edges, and the escalation chain and the risk-tier gating are already configurable inputs on the control type. If the SOC Game finds nothing else, it should confirm this.

escalation

Definition: Handing an alert upward because it exceeds the authority or the scope of the person holding it.

Distinguishes it from: The escalation chain in CHART, which is the configured route the handing-up follows. The chain is the notation. Escalation is the act, and it carries something the chain does not: the judgement that this one is above me, made by the person least equipped to make it.

Operational test: Ask what happens if nobody escalates. In a centre the alert closes as a false positive and nothing marks the error. Under a control edge the action simply does not fire, and the difference is that one of the two failures is silent.

In my project: The silent failure is the interesting one and the graph does not currently have a

place to put it. A control edge models an action waiting for permission. It does not model a decision that was never raised.

playbook

Definition: The written procedure for a class of alert, saying what to check in what order and what to do with each outcome.

Distinguishes it from: A temporal dependency, which is the notation's way of saying that one action must precede another. A playbook is longer than an ordering: it is the ordering plus the branch conditions plus what counts as enough checking, and only the first of the three has an edge type.

Operational test: Try to write the playbook as edges. The steps go in. The conditions under which a step is skipped do not.

In my project: This is where I expect the succinctness failure rather than the completeness one. Everything a playbook says can probably be forced into the graph. Whether the result is readable by the people who have to follow it is a separate question and the one that matters.

alert fatigue

Definition: The degradation in judgement that follows from a volume of alerts beyond what the team can process, showing up as unexamined queues, delayed investigation, and missed detections rather than as visible error.

Distinguishes it from: Workload as it is measured in my own studies, which is asked of a participant after a session with a questionnaire. Fatigue in a centre is a property of a queue over months, and no session-length instrument reaches it.

Operational test: Ask what happened to the alerts nobody opened. If the answer exists and is large, the team is fatigued whatever anyone reports.

In my project: It is the reason human-AI teaming is proposed for centres at all, and it is the outcome variable a SOC Game would most want and least be able to produce. This is the strongest argument that the test has to be run partly by encoding real centres rather than only by running sessions.

shift handover

Definition: The transfer of an open picture between two sets of people at a shift boundary: what is running, what was decided, what was deliberately left.

Distinguishes it from: Everything in the notation, which is the point. A CHART graph is a structure over teammates present in one session. A handover changes who the teammates are without ending the work, and the graph has no construct for the same role held by a different person an hour later.

Operational test: Ask whether the team at the end of the task is the team that started it. In a centre it is not, routinely, and in every study in this cluster it is.

In my project: This is the first place I expect the encoding to fail outright. It is also the most interesting failure, because a construct for it would be a construct for continuity of authority across a change of personnel, which is a general teaming problem and not a cyber one.

incident response team

Definition: The group assembled around a confirmed incident, with a commander who holds decision authority for its duration and specialists who join for the parts that need them.

Distinguishes it from: The centre, in both structure and lifetime. The centre's authority is standing and tiered. The response team's is temporary and centralized in one person, and it is dissolved on purpose when the incident closes.

Operational test: Ask who can be overruled. In a response team, one person overrules everyone. In a centre, authority is distributed by tier and nobody holds all of it.

In my project: My sentence asks about both in one breath, which hides that they are two different tests. The response team is the easier one: a single commander is a control star, and the graph writes it in one line. The centre is the hard one.

1.3 Three things the graph does not yet say

queue depth

Definition: The number of alerts standing unjudged at a moment, and the rate at which the number is changing.

Distinguishes it from: Any quantity currently in a CHART trace. The trace records what the team did. Queue depth is a fact about what the team did not reach, and it cannot be recovered from a record of actions taken.

Operational test: Look for the alert nobody touched. If the record has no entry for it, the record cannot report the load.

In my project: Adding it is not a change to the dependency graph. It is a change to what a session contains: a stream of work arriving faster than it can be cleared, which none of my four studies has.

authority that shifts

“Real teams move authority around as the situation changes, the human takes the wheel when stakes rise, the agent runs ahead when they don’t.”

website, Next, authority that shifts

Definition: Reassignment of who may act without asking, during the task, triggered by the state of the task.

Distinguishes it from: A fixed graph, which is what CHART configures before the session and holds for its duration. My own Next page already names this as the third open question of the cluster, on general grounds. The centre is where it stops being a refinement and becomes the ordinary case.

Operational test: Ask whether the same action needs approval at hour one and at hour three. In a centre under a live incident it does not.

In my project: The Next page asks when the shift should happen, whether it can be learned, and how to allow it without giving up the safety the fixed structure was there to guarantee. The

SOC Game is where that question would be posed against a real answer rather than a designed one.

tacit knowledge in the centre

Definition: What an experienced analyst knows about their own network that is not in any playbook and is transferred, when it is transferred at all, by sitting next to somebody.

Distinguishes it from: The cluster's general use of the term, which is about the human-AI seam and what can be handed across it. Here the seam is between two people, and the field studies are consistent that the transfer happens through proximity and shared simulation rather than through documentation.

Operational test: Ask a senior analyst to write down how they knew. The part they cannot write is the part at issue.

In my project: The cluster's first challenge is that complementarity depends on knowledge neither side can fully state. A centre is the setting where that claim has actually been studied, by interview and by embedded fieldwork, and it is the reason those two field studies are on my reading table rather than my platform papers.

2 Working

SIEM and SOAR

The two pieces of software a centre's workflow is built around. The first collects and correlates events and is what emits the alerts. The second automates the response steps and is where a playbook stops being a document and starts being executable. Any encoding of a real centre has to decide which of the two is a teammate node and which is the environment, and I do not think the answer is obvious.

false positive

An alert that turns out not to be an event. It is the dominant category by volume in every centre studied, and its rate is the parameter that decides how much attention a real alert can be given. The FriendOrFoe design already varies it deliberately, which is the one place in this cluster where the centre's actual constraint appears in a study.

mean time to detect and mean time to respond

The two clocks a centre is managed against. They are the reason a centre will accept an agent that is wrong sometimes and fast always, and they are also why an approval requirement is expensive there in a way it is not in a laboratory session.

threat hunting

Looking for what no detector raised, done by the tier with the most authority and the least procedure. It is the part of the work that has no alert to anchor it, which makes it the part hardest to represent as a response to anything, and therefore hardest to write as edges over a triggering event.

root cause analysis

The step every interviewed analyst converges on once an incident is confirmed: working backward

from the observed effect to the entry point. It is the SOC's own name for what the trace vocabulary in this cluster calls attribution, and the two should be kept apart, because one is done to a network and the other is done to a team.

autonomy level

The framing the operational literature uses in place of a dependency graph: a ladder from an AI that handles routine alerts alone, through an AI that speeds up an expert's decision, to a person and a model working a novel case together, with the task's criticality deciding which applies. It describes the same territory as the control dependency and describes it as a setting on a dial rather than as a structure, which is exactly the difference the SOC Game would put to the test.

3 Peripheral

the field studies

Three of them fix what is known about how centres actually work. Four years of embedded fieldwork across three centres, which produced the account of burnout as a self-reinforcing cycle among skill, discretion and trust, with the most experienced analysts leaving first. Eighteen interviews with analysts and managers, which found the two groups disagree about what the centre is for. Ten interviews on thought process during incidents, which produced the root cause finding and the observation that proximity and simulation move knowledge better than documents do. Two of the three are already on my reading table.

the incident response profile

The federal guidance moved in 2025 from a four-phase incident handling lifecycle to a set of recommendations mapped onto the six functions of the cybersecurity framework, which puts governance and identification alongside detection and response rather than before them. It matters here only as a constraint: a centre being encoded is usually also being audited against this, and the encoding should not need a second vocabulary.

the NSF award

Adversarial Causal Reasoning for SOC, a CyberAI Innovation award to Wajih Ul Hassan, recorded on my Headspace page under the question about what the current SOC stack involves. It is on that page rather than in a project folder because the question it attaches to is mine and the award is not.

4 The disagreements, listed

The disagreement	The entry it sits in
A centre and a response team, asked about in one sentence	incident response team
Escalation as an act, escalation chain as a route	escalation
Workload as a questionnaire, fatigue as a queue over months	alert fatigue
Root cause analysis on a network, attribution on a team	root cause analysis

The first is the one to fix before any design work starts. My own sentence asks whether the graph can express the teamwork in a centre or a response team, and the two are different enough that a single answer would be worthless. A response team is one commander with temporary authority over specialists, and CHART writes it in one line. A centre is standing tiered authority over a queue that never empties, staffed by people who change every eight hours. The graph is very likely to pass the first test and fail the second, and the failure is the result worth having.

5 What crosses into the shared lexicon

Terms this project uses that the whole cluster uses, whose cluster sense sits in the shared file: authority boundary, control dependency, informational dependency, trace, over-reliance, human load, and tacit knowledge. The entries above carry only what a real centre adds to them.

Three terms are fixed here and nowhere else in the cluster. Expressiveness, with completeness and succinctness under it, is the only property in my files that is asserted of a notation rather than of a team. Queue depth is the only quantity in my files that is about work not done. And shift handover is the only teaming arrangement I have named that the dependency graph has no construct for at all.

One term comes in from the cluster's own open questions and should stay attached there. Authority that shifts is the third question on the Next page, and the SOC Game is the setting that would answer it rather than a project that owns it.

6 Sources

- The two statements of the question are from the Human-AI Complementarity overview page and from `content/artifacts/chart/README.md`. The direction of the work and the requirement to model how a real centre divides authority are from the Testing What CHART Can Express posting on Work With Me. The third open question is from the cluster's Next page.
- The SOC stack questions and the NSF award are from the cyber agents and physical agents section of Headspace.
- The five dependency types, the escalation chain and the risk-tier gating are from `chart.md` and `content/artifacts/chart/design.md`, and belong properly to `hac_chart.tex`. They appear

here only where a real centre tests them.

- The false alarm manipulation and the binary triage task are from the FriendOrFoe design and belong to `hac_friendorfoe.tex`.
- The tiering, the alert volume figures and the alert fatigue account are from the operational and survey literature on security operations centres, including the recent computing surveys treatment of alert fatigue and its research challenges.
- The four-year embedded fieldwork and the burnout cycle are from Sundaramurthy et al., A Tale of Three Security Operation Centers, ACM Workshop on Security Information Workers, 2014, and the dissertation behind it. The eighteen interviews and the manager-analyst disagreement are from Matched and Mismatched SOCs, ACM CCS 2019. The ten interviews, the root cause finding and the transfer result are from Cho, Happa and Creese, Capturing Tacit Knowledge in Security Operation Centres, IEEE Access 8, 42021 to 42041, 2020. The first and third are on my reading table.
- The autonomy ladder is from Chhetri, Uzunov, Vo, Nepal and Kowalczyk, Towards Human-AI Teaming to Mitigate Alert Fatigue in Security Operations Centres, ACM Transactions on Internet Technology 24(3), 2024, which sets out the automated, augmented and collaborative modes.
- The move from a four-phase lifecycle to the six framework functions is from NIST SP 800-61 Revision 3, April 2025.
- The three tiers, the three fluency levels and the four fields are from the Tech Fluency page. The reason a term whose two uses disagree belongs inside its entry is in `lexicon_method.tex`, in this folder.